

Technical Document #264: Data Engineering - Comprehensive Overview

Technical Document #264: Data Engineering - Comprehensive Overview

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Data quality ensures data is accurate, complete, and consistent.
- Data pipelines automate the movement and transformation of data between systems.
- Stream processing handles continuous data streams in real-time.